

Green and black tea in relation to gynecologic cancers

Scope Observational studies have evaluated the relationship between green tea intake and cancers of the ovary and endometrium, but we are not aware of the published studies on green tea intake and risk of human papillomavirus (HPV)-related cancers of the cervix, vagina, or vulva. Methods and results A critical review of the published literature on tea intake and risk of ovarian and endometrial cancers was conducted. In meta-analyses, we report inverse associations for green tea intake and risk of ovarian cancer (odds ratio [OR] = 0.66; 95% confidence interval [CI]: 0.54, 0.80), and for green tea and risk of endometrial cancer (OR = 0.78, 95% CI: 0.62, 0.98). There was no association for black tea and ovarian cancer risk (OR = 0.94, 95% CI: 0.87, 1.02) and a positive association with endometrial cancer risk (OR = 1.20, 95% CI: 1.05, 1.38). We summarized the experimental evidence supporting the antiviral and immunomodulatory activities of green tea catechins, and results from randomized clinical trials that demonstrated green tea catechin efficacy on treatment of cervical lesions and external genital warts. Conclusion Observational data support a protective role of green tea on risk of ovarian and endometrial cancers. Observational data are needed to evaluate whether green tea reduces risk of human papillomavirus-related cancers.